

Accurate prediction of synthesizability and precursors of 3D crystal structures via large language models

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Accessing the synthesizability of crystal structures is crucial for transforming theoretical materials into real-world applications. Nevertheless, there is a significant gap between actual synthesizability and thermodynamic or kinetic stability commonly used to screen synthesizable structures. Herein, we develop the Crystal Synthesis Large Language Models (CSLLM) framework, which utilizes three specialized LLMs to predict the synthesizability of arbitrary 3D crystal structures, possible synthetic methods, and suitable precursors, respectively. We construct a comprehensive dataset including synthesizable/non-synthesizable crystal structures and develop an efficient text representation for crystal structures to fine-tune LLMs. Our Synthesizability LLM achieves state-of-the-art accuracy (98.6%), significantly outperforming traditional synthesizability screening based on thermodynamic and kinetic stability. Its outstanding generalization ability is further demonstrated in experimental structures with complexity considerably exceeding that of the training data. Furthermore, both the Method and Precursor LLMs exceed 90% accuracy in classifying possible synthetic methods and identifying solid-state synthetic precursors for common binary and ternary compounds, respectively. Leveraging CSLLM, tens of thousands of synthesizable theoretical structures are successfully identified, with their 23 key properties predicted using accurate graph neural network models.

The journey of materials design has undergone a remarkable evolution, characterized by four paradigms¹. While the first and second paradigms built the foundation through trial-and-error experiments and scientific theories, respectively, the third paradigm leveraged computational methods, such as density functional theory (DFT) for efficient atomic-level simulations to identify numerous functional materials^{2,3}. The fourth paradigm represents a transformative leap by harnessing the power of accumulated data and machine learning (ML), significantly accelerating materials discovery by predicting properties rapidly and accurately^{4–13}. Notably, generative models within this paradigm facilitate the inverse design of novel material structures^{14–17},

such as semiconductors, zeolites and metal-organic frameworks^{18–20}, overcoming the limitation of traditional ML restricted to known structures. While the computational and data-driven paradigms have collectively identified millions of candidate materials with excellent properties, a great challenge remains—the synthesizability of predicted material structures, which is imperative for transforming theoretical innovations into real-world applications.

Conventional approaches for identifying promising synthesizable material structures typically involve assessing thermodynamic formation energies or energy above convex hull via DFT calculations^{16,17,21,22}. However, numerous structures with favorable formation energies have

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yet to be synthesized²³, while various metastable structures are synthesized with less favorable formation energies at the convex hull minimum²⁴. An alternative approach involves assessing kinetic stability by computationally expensive phonon spectra analyses²⁵. Nonetheless, material structures with imaginary phonon frequencies can still be synthesized²⁶. Jones et al. combined basin hypervolume with thermodynamic stability to explain the synthesis of metastable silicon polymorphs²⁷. Phase diagrams offer a more direct correlation with synthesizability, as they delineate stable phases under varying temperatures, pressures, and compositions^{28,29}. However, constructing the free energy surface for all possible phases as a function of these variables is computationally impractical. Moreover, the synthesis of materials is a complex process influenced by the choice of precursors and reaction conditions³⁰. Thus, there is an urgent need for a method that can accurately and rapidly predict synthesizability and suitable precursors, thereby providing direct guidance for experimental synthesis.

ML has shown promising attempts in predicting material synthesizability, such as the SynthNN model for assessing the synthesizability of inorganic crystals based on their compositions³¹. When predicting the synthesizability of crystal structures, constructing negative samples (non-synthesizable materials) is a major challenge. The simplest approach treats structures with unknown synthesizability as negative samples, which inevitably introduces numerous synthesizable structures³². Another method collects unobserved structures from well-studied compositions, yet such datasets are limited in scale (only 3000 samples)³³. More commonly, semi-supervised methods like positive-unlabeled (PU) learning are used—achieving over 75% and 87.9% accuracy in synthesizability prediction for 2D MXenes³⁴ and 3D crystals³⁵, respectively. Additionally, the teacher-student dual neural network further improved the prediction accuracy for 3D crystals to 92.9%³⁶. Notably, our previous work utilized failed experimental data as negative samples, though constrained to 2D silver/bismuth organic-inorganic hybrid perovskites¹².

Overall, these ML methods for synthesizability prediction exhibit moderate accuracy or are confined to specific material systems, while also lacking the capability to simultaneously predict synthesis routes and precursors. Recent advances in large language models (LLMs), such as ChatGPT³⁷ and open-source LLMs like LLaMA³⁸, have revolutionized various research domains. In materials science, LLMs facilitate predicting material properties³⁹, optimizing experimental workflows⁴⁰, generating crystal structures⁴¹, proposing synthesis strategies for organic molecules⁴², metal-organic frameworks⁴³, and evaluating synthesizability based on inorganic material formulas⁴⁴. These demonstrate the exceptional capabilities of LLMs to learn from text due to their extensive architectures and vast training datasets. Thus, LLMs hold great potential to predict the synthesizability of theoretical crystal structures. However, challenges persist in curating data to fine-tune LLMs for ensuring accuracy and reliability. Crystal structures exhibit vast diversity, yet the available data (10^5 – 10^6) pales in comparison to that of organic molecules (10^8 – 10^9)^{45,46}. Meanwhile, there is an absence of an effective text representation method for crystal structures, akin to the SMILES⁴⁷ notation for organic molecules. Additionally, the issue of “hallucination” in LLMs, where the generated information is not grounded in real-world facts⁴⁸, poses a risk to the reliability of synthesizability predictions.

In this work, we address the prediction of crystal structure synthesizability into three tasks: i) establishing whether a structure is synthesizable, ii) pinpointing the appropriate synthetic method (solid-state or solution), and iii) identifying the appropriate precursors. These challenges are addressed using three fine-tuned LLMs, which collectively constitute a framework denoted as Crystal Synthesis Large Language Models (CSLLM), as illustrated in Fig. 1a. We curated a balanced dataset containing 70,120 synthesizable crystal structures from the Inorganic Crystal Structure Database (ICSD)⁴⁹ and 80,000

non-synthesizable structures screened from 1,401,562 theoretical structures via a PU learning model³⁵. To facilitate efficient LLM fine-tuning, we introduced a text representation termed “material string” that integrates essential crystal information. The Synthesizability LLM achieves a remarkable 98.6% accuracy on testing data, outperforming thermodynamic method—energy above hull ≥ 0.1 eV/atom (74.1%) and kinetic method—lowest frequency of the phonon spectrum ≥ -0.1 THz (82.2%). Moreover, it demonstrates exceptional generalization by predicting the synthesizability of additional testing structures, achieving 97.9% accuracy even for complex structures with large unit cells. The Method and Precursor LLMs achieve 91.0% classification accuracy and 80.2% precursor prediction success. We also calculated reaction energies and performed a combinatorial analysis to suggest more potential precursors. Furthermore, CSLLM was used to assess the synthesizability of 105,321 theoretical structures, and 45,632 synthesizable materials were screened out, whose 23 key properties were further predicted in batches by GNNs. Finally, A user-friendly CSLLM interface was developed for automatic synthesizability and precursor predictions from uploaded crystal structure files (Fig S1). These achievements arise from domain-focused fine-tuning, which aligns the broad linguistic features of LLMs with material features critical to synthesizability, thereby refining its attention mechanisms and reducing hallucinations. Such domain adaptation has been widely reported to yield substantial performance boosts for LLMs even when only relatively small datasets are available^{39,50}. This groundbreaking framework bridges the gap between theoretical predictions and practical synthesis, thereby paving the way for the efficient discovery of novel materials.

Results

Balanced and comprehensive dataset for predicting synthesizability

To develop a robust LLM for predicting the synthesizability of 3D crystal structures, it is essential to construct a comprehensive dataset comprising synthesizable (positive examples) and non-synthesizable (negative examples) materials. The ICSD⁴⁹ serves as a reliable source of experimentally validated material structures, all of which are confirmed to be synthesizable. For our study, we meticulously selected 70,120 crystal structures from the ICSD, each containing no more than 40 atoms and seven different elements, as the positive examples. Notably, disordered structures within the ICSD were excluded from our selection, as our objective focuses on the prediction of the synthesizability of ordered crystal structures.

To construct a set of negative examples, we employed a pre-trained PU learning model developed by Jang et al.³⁵. This model generates a CLscore for each structure, with scores below 0.5 indicative of non-synthesizability. This enables us to discern non-synthesizable structures from a vast pool of 1,401,562 crystal structures sourced from the materials project (MP)⁵¹, Computational Material Database⁵², Open Quantum Materials Database⁵³, and Joint Automated Repository for Various Integrated Simulations (JARVIS)⁵⁴. In order to build a balanced dataset, we calculated the CLscores for all 1,401,562 structures and selected the 80,000 structures with the lowest CLscores (*i.e.*, CLscore < 0.1) as our non-synthesizable examples (Fig. 1d). Note that we computed the CLscores for our 70,120 positive examples and 98.3% of these structures had CLscores greater than 0.1, which affirmed the validity of our CLscore threshold.

All 150,120 crystal structures are visualized using t-SNE⁵⁵, covering seven crystal structures: cubic, hexagonal, tetragonal, orthorhombic, monoclinic, triclinic, and trigonal with the cubic system being the most prevalent (Fig. 1b). Additionally, our dataset includes crystal structures with 1–7 elements, predominantly featuring 2–4 elements, and covers atomic numbers 1–94 from the periodic table, excluding atomic numbers 85 and 87 (Fig S2). This demonstrates that our dataset is both balanced and comprehensive, laying a solid foundation for the

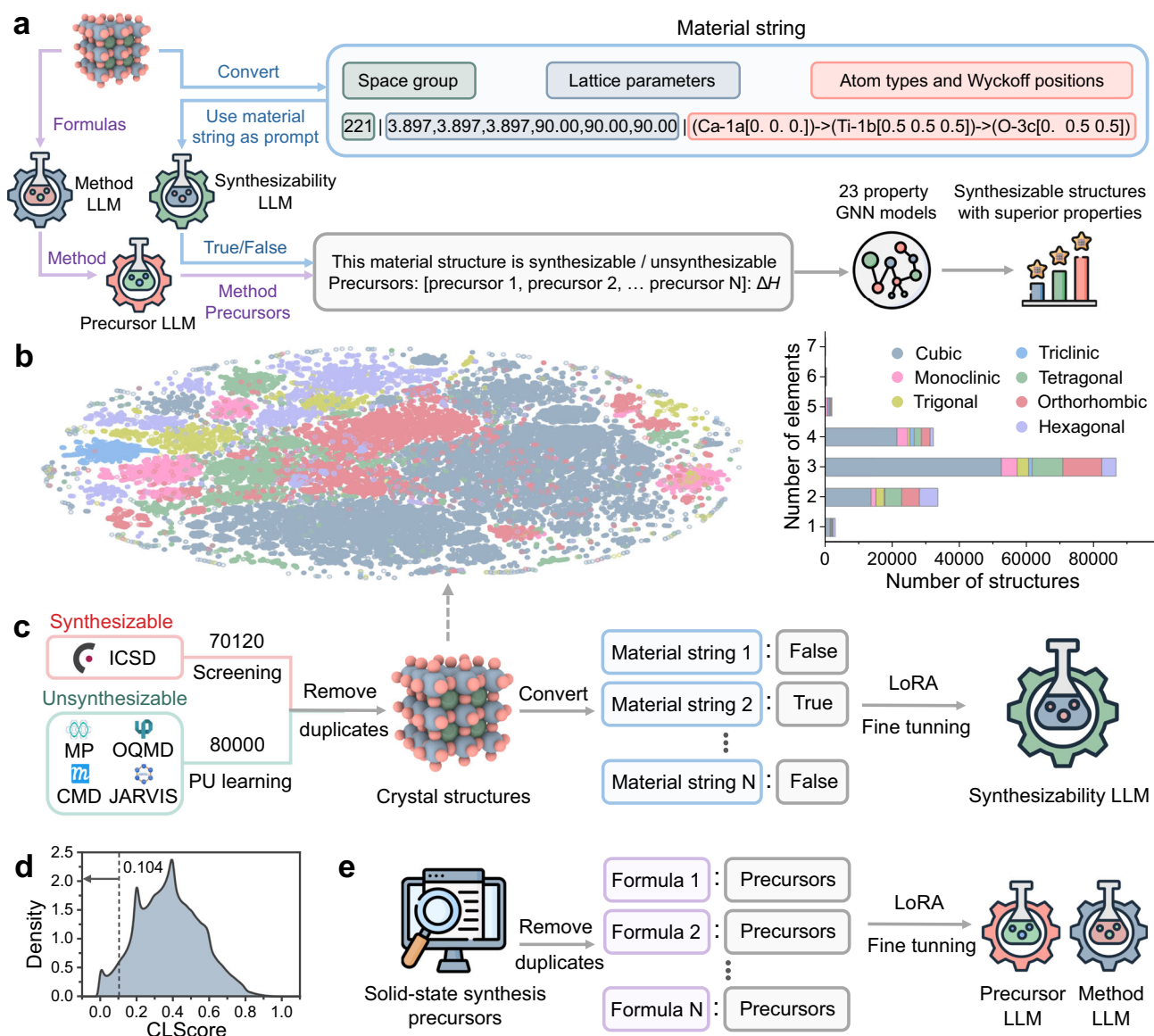


Fig. 1 | Crystal Synthesis Large Language Models (CSLLM) framework and data for fine-tuning large language model (LLM). **a** Overall workflow for predicting the synthesizability and recommending precursors using the LLM. Crystal structures were first converted into material strings. Their synthesizability was then predicted by the Synthesizability LLM. Based on the chemical formulas, the Precursor and Method LLMs provide a batch of potential precursors along with their reaction energies. 23 material properties of synthesizable structures were further predicted by graph neural networks. **b** t-distributed Stochastic Neighbor Embedding (t-SNE) visualization of 150,120 material structures along with statistical data on crystal systems and the number of elements, which cubic, hexagonal, tetragonal, orthorhombic, monoclinic, triclinic, and trigonal structures represented by gray, purple,

cyan, red, orange, blue, and green dots, respectively. **c** Training process of the Synthesizability LLM. A balanced dataset was constructed from Inorganic Crystal Structure Database (ICSD)⁴⁹, Materials Project (MP)⁵¹, Open Quantum Materials Database (OQMD)⁵³, Computational Material Database (CMD)⁵², and JARVIS (Joint Automated Repository for Various Integrated Simulations)⁵⁴ using a positive-unlabeled learning model, converted to material strings in batches, and then fine-tuned LLMs with low-rank adaptation (LoRA). **d** Crystal-likeness score (CLScore) distribution for 1,401,562 structures and dash line refers to the CLScore threshold for filtering out non-synthesizable structures. **e** Training process for the Precursor and Method LLMs. Data pairs of chemical formulas with their precursors and synthetic methods were collected from the literature and fine-tuned with LoRA.

subsequent training of a high-fidelity LLM for predicting the synthesizability of inorganic crystal structures.

Text representation of material structures and fine-tuning of synthesizability LLM

Given that LLMs process text inputs, it is essential to represent material structures in the simplest reversible text format that includes comprehensive information on the lattice, composition, atomic coordinates, and symmetry. The most common text representations of crystal structures are the CIF format⁵⁶ and the POSCAR format used by the Vienna Ab initio Simulation Package⁵⁷. These formats provide detailed

information on the lattice, composition, and atomic coordinates, but contain redundant information. For example, multiple atomic coordinates at the same Wyckoff position can be inferred from one atomic coordinate along with the space group and Wyckoff position symbols. Therefore, it is unnecessary to list all the atomic coordinates within the cell. Additionally, POSCAR format lacks symmetry-related information, although it is more concise than CIF. To address these issues, we propose a text representation for crystal structures named material string: $SP | a, b, c, \alpha, \beta, \gamma | (AS_1-WS_1[WP_1]) \rightarrow (AS_2-WS_2[WP_2]) \rightarrow (AS_N-WS_N[WP_N])$, where the SP, (a, b, c, α , β , γ), AS, WS and WP are the space group number, lattice constants, atom symbol, Wyckoff position symbol and

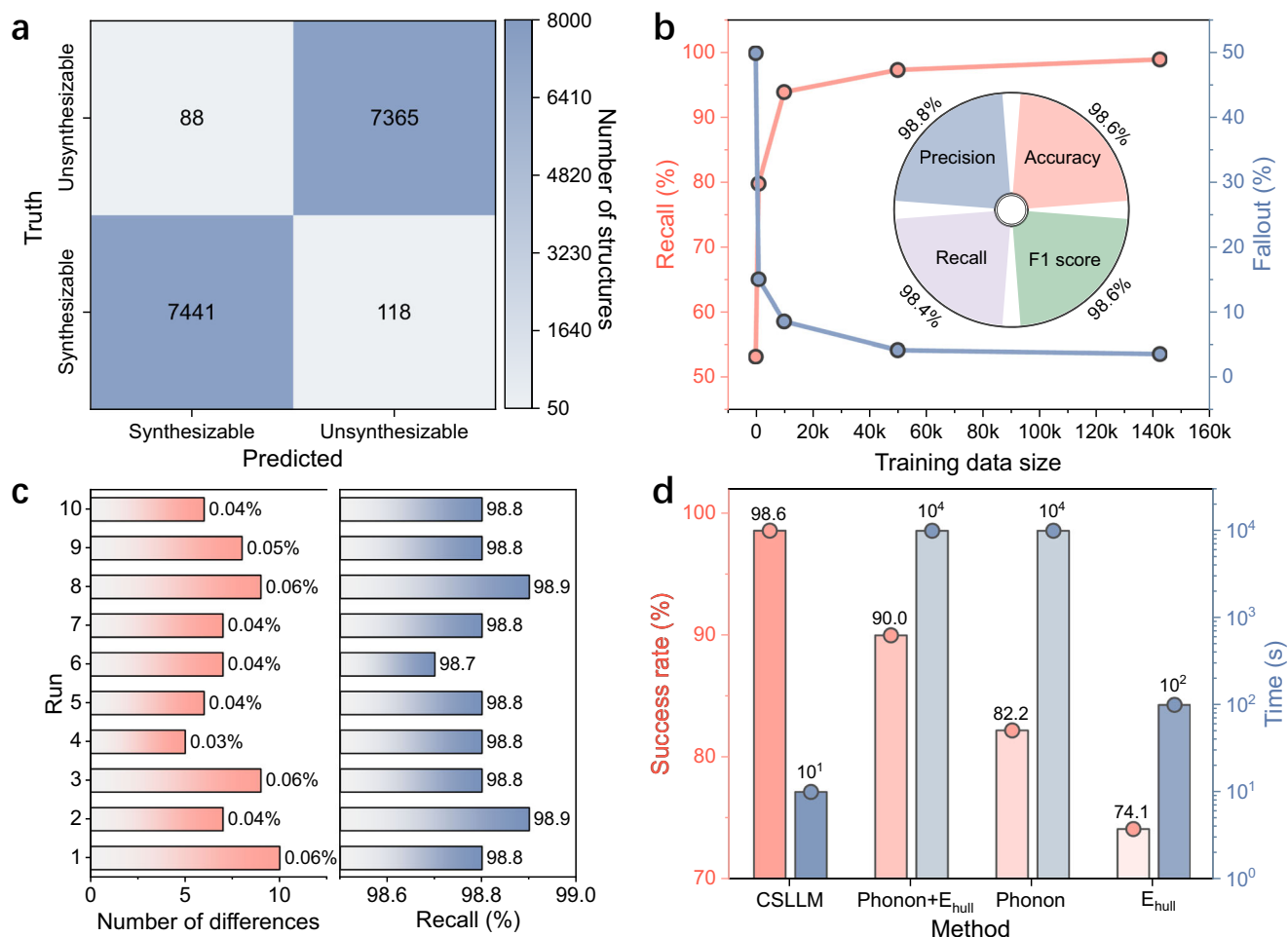


Fig. 2 | Performance of Synthesizability large language model (LLM).

a Confusion matrix of the Synthesizability LLM on the test set. **b** Recall and fallout of the Synthesizability LLM under different amounts of training data. The pie chart subfigure shows the final classification metrics of the model trained with the full dataset. **c** Variations in results and recall of Synthesizability LLM for ten additional test set inferences (Runs). The left subplot shows the number and percentages of

differences between each run and initial inference results, while the right subplot displays the recall value of each run. **d** Comparison of the accuracy and runtime of synthesizability predictions for crystal structures between the Synthesizability LLM and density functional theory computational methods (E_{hull} refers to the energy above hull).

Wyckoff position of a crystal structure, respectively. For instance, the material string of the CaTiO_3 structure (mp-4019) from MP database is “221|3.897.3.897.3.897.90.00.90.00.90.00|(Ca-1a[0.0.0.])>(Ti-1b[0.5 0.5 0.5])>(O-3c[0.5 0.5 0.5])”(Fig. 1a). Notably, crystal structures are first converted into primitive cells before converting to material strings.

The material string not only contains all the crystallographic information but also is as concise as possible. For example, the converted CaTiO_3 structure has a string length of only 102 characters, significantly shorter compared to 1817 characters in CIF format and 1644 characters in POSCAR format. This high-information-density text is beneficial for improving the fine-tune performance of LLMs⁵⁸. Thus, we converted 150,120 unique crystal structures and their synthesizability labels into material strings–True/False data pairs, then split them into training and testing sets in a 9:1 ratio. Notably, due to the deduplication of crystal structures, the material strings in the testing set did not appear in the training set. Some examples of these data pairs are shown in Fig S3. Considering the substantial expenses associated with the comprehensive fine-tuning of LLMs, we utilized the low-rank adaptation (LoRA) method⁵⁹ to efficiently refine the LLaMA3-8B model. This approach enables cost-effective tuning by targeting the optimization of a limited subset of parameters.

The fine-tuned synthesizability LLM achieves remarkably high accuracy on the test set, with only 88 incorrect predictions out of

7529 unseen synthesizable crystal structures (Fig. 2a). To demonstrate that the outstanding ability of LLM to distinguish synthesizable crystal structures is a result of our fine-tuning rather than inherent to the model, we trained the synthesizability LLM with 0, 1000, 10,000, 50,000, and 135,108 data pairs for comparison. As shown in Fig. 2b, the recall (the proportion of actual positives correctly identified) and fallout (the proportion of actual negatives incorrectly identified as positive) of the untuned LLM are only 53.1% and 49.0%, respectively, almost equivalent to random guessing. After fine-tuning with 1000 data points, the recall and fallout of synthesizability LLM improve to 79.8 and 9.36%, demonstrating the powerful ability of LLMs to learn patterns from textual data. Ultimately, the precision, accuracy, recall, and F1 score reach 98.8, 98.6, 98.4, and 98.6%, respectively. The detailed testing performance of the five LLMs in Fig. 2b are shown in Table S1 and Fig S4. Furthermore, we interpret the synthesizability LLM using perturbation-based feature ablation method, revealing the most contributive features (i.e., space group number or lattice constants or atom symbol or Wyckoff position) in the prompts (material string) for synthesizability prediction (details are discussed in Supplementary Note 1 and Fig S5–9). “This function is incorporated into the CSLLM framework, the importance score of each feature is automatically calculated after synthesizability prediction.”

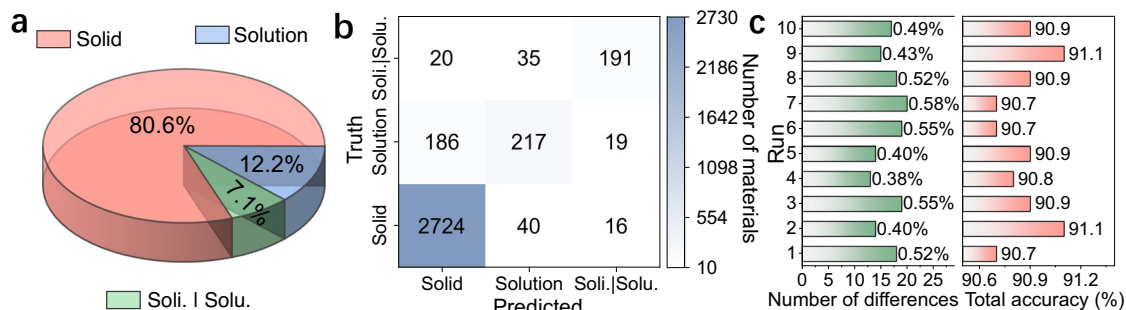


Fig. 3 | Performance of Method large language model (LLM). **a** Proportion of solid-state, solution, and dual-method synthesis among the 31,780 synthesis data points. “Soli.|Solu.” refers to “Solid|Solution” (i.e., the combined methods).

b Confusion matrix of the Methods LLM on the test set. **c** Variations in results and accuracy of the Methods LLM for ten additional test set inferences compared to the initial results.

However, identical prompt input into LLMs often yields different or even erroneous results due to the inherent hallucination problem present in LLMs. Hallucination, defined as the generation of plausible yet incorrect or nonsensical information, is a well-documented issue in LLMs, affecting the reliability across various applications³⁶. To demonstrate that our synthesizability LLM can reliably predict the synthesizability of crystal structures without hallucination, we conducted an additional ten inference runs on the test set. As shown in Fig. 2c, the largest discrepancy in synthesizability predictions compared to the first run is only 10 instances, representing 0.06% of the total. The recall rate varies by no more than 0.2%, affirming the stability of the synthesizability LLM.

Building on this stability, we further compared the synthesizability LLM with three DFT-based methods commonly used for screening experimentally synthesizable material structures: (i) energy above hull (E_{hull}) ≥ 0.1 eV/atom, (ii) the lowest frequency of the phonon spectrum (Phonon) ≥ 0.1 THz, and (iii) both $E_{\text{hull}} \geq 0.1$ eV/atom and Phonon ≥ 0.1 THz. This comparison utilized data from 1512 structures in the MP database for which phonon spectrum calculations had been conducted. The E_{hull} , the lowest phonon frequency, and synthesizable/non-synthesizable labels of these 1512 structures are collected. The distribution of the lowest phonon frequency is shown in Fig S10, where 34.2% of these structures have imaginary frequencies. We then performed inferences on these structures using the synthesizability LLM and assessed the accuracy of the aforementioned methods based on the proportion of experimentally synthesized structures that met the above criteria. As shown in Fig. 2d, the synthesizability LLM not only achieves a significantly higher accuracy of 98.6% compared to DFT methods, but also reduces runtime by 1–3 orders of magnitude, demonstrating its capability to accurately and rapidly predict the synthesizability of 3D crystal materials.

Fine-tuning of method and precursor LLMs

After identifying synthesizable material structures, suitable synthesis recipes (methods and precursors) need to be selected before conducting experimental synthesis. The prevalent synthetic methods encompass the solid-state and solution method, each requiring different precursors. To build a comprehensive dataset for synthesis recipes prediction, we integrated data from 19,488 solid-state synthesis recipes collected by Kononova et al.⁶⁰ and 35,675 solution synthesis recipes collected by Wang et al.⁶¹. Additionally, we extracted 6,136 solid-state/solution synthesis recipes from the literature spanning 2021–2024, utilizing scripts provided by Kononova et al.⁶⁰. Collectively, these data link each chemical formula to a specific synthetic method and a set of precursors, with a subset also containing structural information. After deduplication, we identified 31,780 unique chemical formula-method/precursors data pairs, which cover a wide range of elements with atomic numbers from 1 to 98 (Fig S12). All these

unique synthesis data were utilized for the subsequent LLM fine-tuning, rather than the smaller subset with structural information, as the latter only has 3650 unique structure-method/precursor data pairs, which are inadequate for effective fine-tuning of LLM (Fig S13).

To anticipate potential precursors, it is imperative to first ascertain the applicable synthetic methodology for a given chemical formula. Thus, the LLaMA3-8B model was initially fine-tuned using the 31,780 formula-method pairs from the abovementioned dataset. The dataset distribution includes solid-state, solution, and combined methods, which account for 80.64, 12.22, and 7.14% of the entries, respectively, as illustrated in Fig. 3a. 90% of the dataset was employed for fine-tuning, with the residual 10% reserved for testing purposes. The Method LLM achieves an overall classification accuracy of 91.0% on the test dataset, significantly outperforming the pre-fine-tuned LLM, which achieves only 38.8% (Fig S14a). However, the prediction accuracy varies across different synthesis methods due to the imbalanced distribution of the dataset. Specifically, as shown in Fig. 3b, the model performs well in predicting solid-state synthesis (97.98% accuracy) but exhibits lower accuracy for solution-based synthesis (51.42%) and combined methods (77.65%). This discrepancy is attributed to the limited training data of the solution and combined methods, highlighting the influence of data availability on model performance. Furthermore, this Method LLM exhibits consistency across 10 inference trials, with the largest discrepancy observed in 18 instances (0.58%) and a maximum fluctuation in the accuracy of 0.4%, as depicted in Fig. 3c.

Next, we split the 31,780 formula-precursors data pairs into training and test sets in a 9:1 ratio and fine-tuned another LLaMA3-8B model to predict possible precursors for arbitrary chemical formulas derived from crystal structures. The prediction accuracy for the solid-state method reaches 80.2% (Fig. 4a), significantly higher than the pre-fine-tuned accuracy of 5.7% on the same test set (Fig S14b). However, the accuracy for the solution method is only 39.7%. Thus, we focused on predicting precursors of solid-state synthesis and conducted an in-depth analysis of the LLM prediction of solid-state precursors. In the test set, discrepancies between predicted and actual precursors fell into two cases (Fig. 4a): (i) cases of missing or additional precursors (6.7%), and (ii) cases where the number of precursors is correct but includes incorrect ones (13.1%). In case (i), most (63.0%) predictions miss one precursor (Fig. 4b). In case (ii), most (73.5%) errors involve a single incorrect precursor (Fig. 4c). Furthermore, we analyzed the above two cases and overall success rates by the number of elements, as shown in Table 1 and Fig S15–20. Notably, the success rate of common binary and ternary materials exceeds 91%, with cases of precursor prediction errors below 5%. For compounds with 4–7 elements, the success rate is relatively low (76.5, 68.4, 56.8, and 50.0%, respectively) due to limited training data; however, most precursors are mis-predicted by just 1–2 incorrect precursors. It is noteworthy that

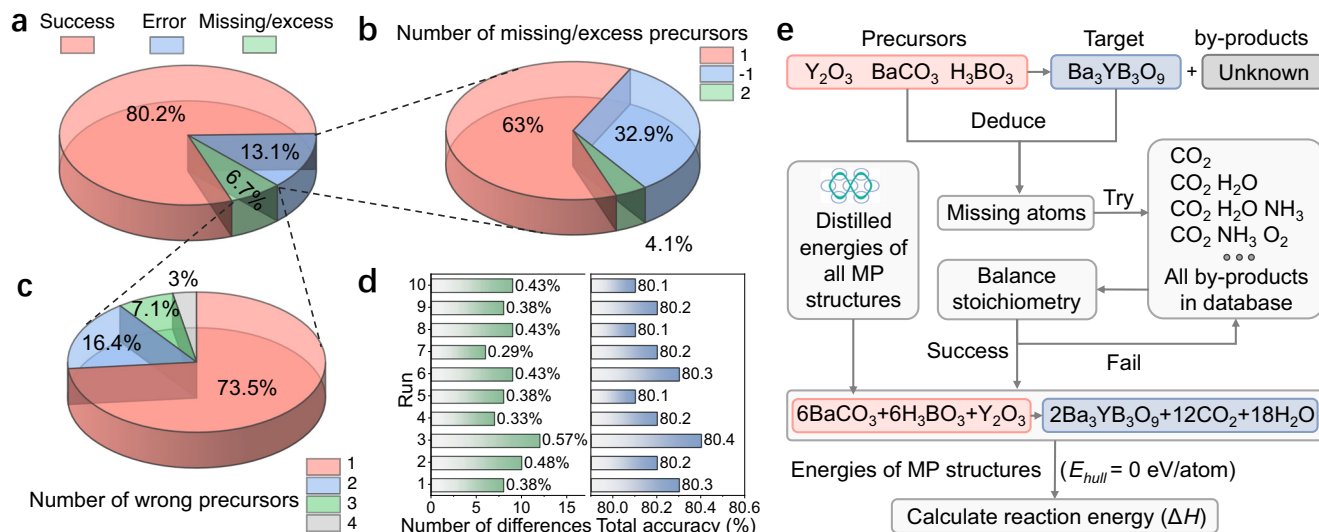


Fig. 4 | Performance of Precursor large language model (LLM). **a** Proportions of precursor predictions by the Precursors LLM that are entirely correct, have missing/excess precursors, or contain errors. **b** Statistics on the number of cases with missing/excess precursors. **c** Statistics on the number of cases with incorrect

precursors. **d** Variations in results and success rate of the Methods LLM for ten additional test set inferences compared to the initial results. **e** Workflow for inferring byproducts and calculating reaction energies based on precursor predictions from Precursors LLM.

the success rate of compounds composed of three, five, six, or seven elements exhibits a linear correlation with the size of the training dataset. In contrast, binary and quaternary compounds do not conform to this linear relationship (Fig S21). Due to the limited dataset of septenary compounds (90 samples), which lacks representativeness, we exclude them from our analysis. This discrepancy may be attributed to the relatively simple precursor composition of binary compounds (typically involving only two types), whereas quaternary compounds often derive from a more diverse array of precursors (generally ranging from three to five types). Moreover, this Precursor LLM passed ten stability tests, with a maximum discrepancy of only 0.57% in inference results and a maximum success rate variation of 0.3% (Fig. 4d). The above analysis demonstrates that our Precursor LLM can provide accurate and robust precursor suggestions.

By combining the fine-tuned method and precursor LLMs, the precursors of solid-state synthesis can be recommended based on the chemical formulas of materials that are identified as synthesizable. To further improve the reliability and practical applicability of our approach, we aimed to calculate the reaction energies using the MP database and explored alternative precursor sets. However, current reaction equations only include the target materials, missing byproducts such as CO_2 and H_2O . To address this, we collected 30 unique byproduct sets from 31,780 material synthesis recipes and iteratively tested these sets to balance the stoichiometry of reaction equations. Next, to calculate the reaction energies, we extracted the total energies of the most stable structures corresponding to the chemical formulas

in the reaction equations from the MP database (Fig. 4e). Furthermore, all possible alternative precursor sets were identified by permuting and combining elements from the initial precursor suggestions (e.g., Y_2O_3 could be replaced with YO_2 or Y and O_2). These combinations were rebalanced and their reaction energies were recalculated (Fig S22). This comprehensive approach broadens the range of potential synthetic routes, making the model more reliable and applicable for real-world synthesis design. Finally, the whole workflow of CSLLM is fully automated and integrated within a graphical user interface (Fig S1), users can simply upload a structure file (CIF or POSCAR format) or type in material string to predict the synthesizability, possible synthetic method, and precursors. The GUI usage example is shown in <https://github.com/szl666/CSLLM>.

Experimental structure validation and identification of synthesizable material structures with outstanding properties

Above, we have demonstrated that CSLLM is capable of accurately and rapidly assessing the synthesizability of material structures in the training set and in-domain test set, as well as identifying the precursors required for synthesis. However, these models lacked experimental validation to confirm their generalization capability. Additionally, our synthesizability LLM was trained on crystal structures with cells containing no more than 40 atoms and seven elements, whereas large materials databases like the MP contain structures exceeding this limit. Therefore, we sought to address these limitations by expanding the application of our synthesizability model to evaluate 20,237 experimentally synthesized structures in the MP database outside the training data. The predictive accuracy for structures with numbers of atoms ranging from 1 to 275 atoms remained consistently high, exceeding 95% across all sizes (Fig. 5a). Notably, the accuracy did not decline for larger structures (more than 100 atoms). With an average accuracy of 97.8%, the nearly uniform accuracy distribution underscores the remarkable generalization capability of the synthesizability LLM even when applied to structures significantly exceeding the complexity of the training set. Moreover, A previous study warned of spurious high accuracy of synthesizability prediction due to large lattice volume bias when using different data sources⁶². However, our crystal structure screening prevented such significant bias, and the synthesizability LLM consistently maintained above 93% accuracy across all lattice volumes (details in Supplementary Note 3 and Fig S23a-b).

Table 1 | Success rate, error rate, and missing/excess rate of the solid-state precursors large language model by number of elements

Number of elements	Success rate	Error rate	Missing/excess rate
All	80.2%	13.1%	6.7%
2	92.2%	4.4%	3.3%
3	91.4%	4.6%	4.0%
4	76.5%	14.1%	9.4%
5	68.4%	24.9%	6.8%
6	56.8%	37.8%	5.4%
7	50.0%	33.3%	16.7%

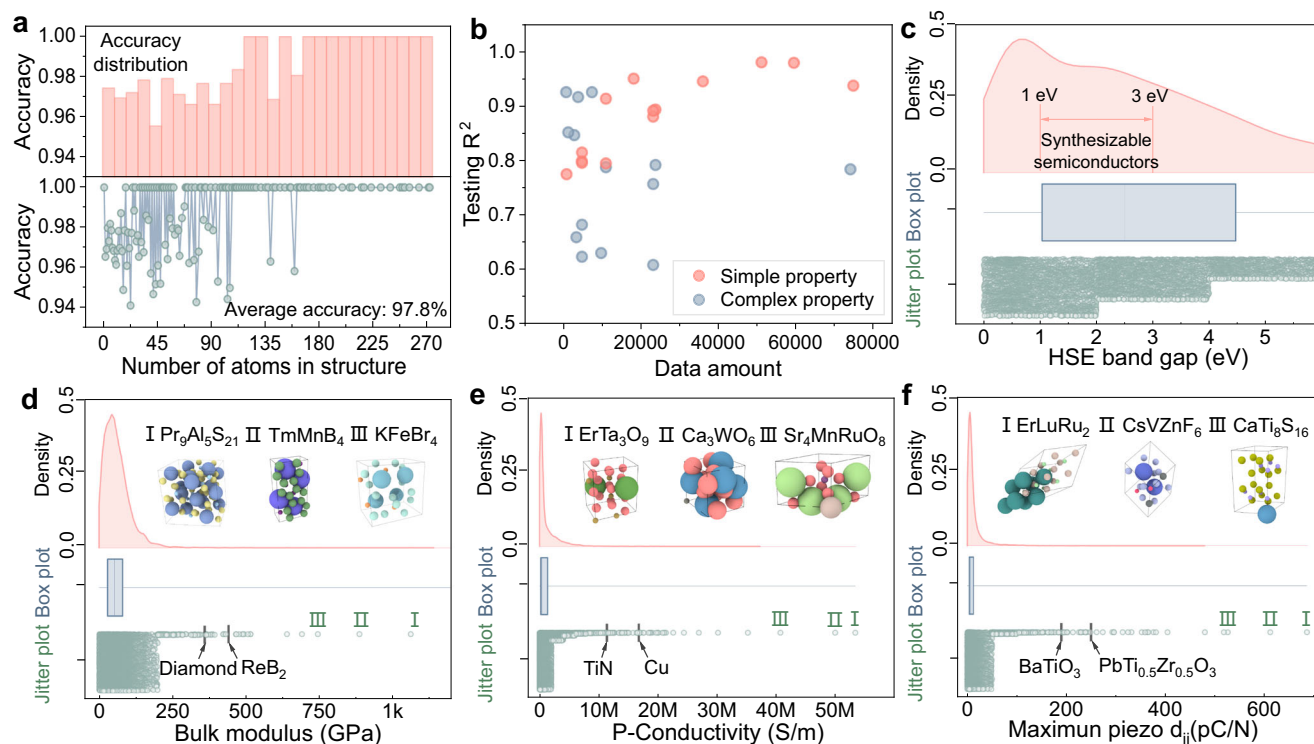


Fig. 5 | Experimental structure validation and property predictions of synthesizable material structures. **a** Synthesizability prediction accuracy for structures with varying numbers of atoms (1–270). The upper plot shows the distribution of prediction accuracy with an interval of 10, while the lower plot depicts the prediction accuracy corresponding to each atom count. **b** Training data quantity vs. test set coefficient of determination (R^2). Red and blue represent simple and complex properties, respectively. **c–f** Kernel Density Estimation (KDE) distribution, box plot, and jitter plot of predicted properties are shown with pink (top), blue (middle), and green (bottom), respectively. In all box plots (**c–f**), bounds of box represent 25–75th percentiles, and whiskers extend to minima and maxima values. For Heyd-Scuseria-Ernzerhof (HSE) band gap (**c**), minima = 0 eV, maxima = 44.12 eV,

median = 2.50 eV, 25–75th percentile = 1.03–4.47 eV; For bulk modulus (**d**), minima = 0.04 GPa, maxima = 3273.62 GPa, median = 52.28 GPa, 25–75th percentile = 29.18–80.73 GPa; For p-type conductivity (**e**), minima = 5.37 $\times 10^7$ S/m, maxima = 5.37 $\times 10^7$ S/m, median = 3.68 $\times 10^5$ S/m, 25–75th percentile = 1.99 $\times 10^5$ –1.36 $\times 10^6$ S/m; For maximum piezoelectric coefficient (**f**), minima = 0.001 pC/N, maxima = 686.50 pC/N, median = 6.78 pC/N, 25–75th percentile = 3.82–12.77 pC/N. Data are presented as individual data points in the jitter plots ($n = 45,632$ synthesizable materials per property, representing independent predictions). No technical replicates were used in the generation of these statistics. Top three material structures are marked by gray short lines.

Building on the remarkable generalization capability of synthesizability LLM across diverse theoretical structures, we definitely want to quickly predict key properties of synthesizable theoretical structures in existing material databases. Thus, we collected 105,321 theoretical structures from the MP database without experimental synthesis labels. Of these, 5,529 structures were removed because they were part of the training set used to fine-tune the Synthesizability LLM, leaving 99,792 structures for synthesizability prediction. Among these, 45,632 structures were identified as synthesizable. We then collected data on 23 material properties, ensuring that each dataset is exclusively derived from either theoretical calculations or experimental measurements, with no intermixing between the two sources. These data include optical, thermodynamic stability, hydration stability, mechanical, optoelectronic, magnetic, topological, thermoelectric, piezoelectric, dielectric, and superconducting properties. For each of these properties, a high-precision graph neural network (GNN) model based on transformer proposed in our previous work, CGTNet⁶³, was trained.

The testing errors (10% data) and data quantities for 23 property prediction models are shown in Table 2 and Fig. 5b. For simple properties that can be directly derived from energy and electronic structure calculations (red points in Fig. 5b), models with larger training datasets exhibit high accuracy. In contrast, complex and experimental properties do not display this trend, likely due to variations in data quality, such as differing experimental protocols used for testing experimental band gap, which limit data comparability.

Next, these GNN models were employed to batch predict the 23 properties of the 45,632 synthesizable theoretical structures. Four of these predicted material properties—HSE band gap, bulk modulus, p-type conductivity and maximum piezoelectric coefficient—are visualized in Fig. 5c–f. 14,097 synthesizable theoretical structures with HSE band gaps within the semiconductor range (1–3 eV) were identified from 45,632 structures (Fig. 5c). For the remaining three properties, while the majority of synthesizable structures do not exhibit remarkable performance, several structures with exceptional properties are uncovered. This subset includes numerous structures predicted to surpass the performance of existing materials with exceptional properties, such as ultra-hard materials diamond⁶⁴ and ReB₂⁶⁵, conductive materials TiN⁶⁶ and Cu⁶⁴, and piezoelectric materials BaTiO₃⁶⁷ and PbTi_{0.5}Zr_{0.5}O₃⁶⁸. The top three structures for each of these properties are featured in Fig. 5b–g. A comprehensive list of all predicted properties for the synthesizable theoretical structures is provided in Supplementary Data 1.

Discussion

To sum up, we have developed an innovative framework—CSLLM, integration of Synthesizability, Method, and Precursor LLM, to predict the synthesizability, synthetic method and precursors of crystal structures, respectively. The Synthesizability LLM, fine-tuned on a comprehensive and balanced material structure dataset, demonstrates state-of-the-art 98.6% accuracy, outperforming conventional thermodynamic and kinetic stability approaches. Notably, its exceptional

Table 2 | Properties, testing mean absolute error (MAR) and data amount of 23 GNN models

Property	Testing MAE	R ²	Data amount
mBJ band gap	0.270 eV	0.952	18,172
HSE band gap	0.380 eV	0.927	7376
Experiment band gap	0.422 eV	0.848	2808
PBE band gap	0.270 eV	0.913	74,992
Electrochemical stability ΔG_{pbx}	0.276 eV	0.918	3820
Energy above hull	0.102 eV	0.981	59,635
Bulk modulus	9.908 GPa	0.895	23,824
Shear modulus	9.933 GPa	0.793	23,824
Maximum phonon frequency	68.14 cm ⁻¹	0.853	1265
Exfoliation energy	37.49 meV/atom	0.776	813
Poisson ratio	0.087	0.796	10,987
SLME	4.902 %	0.631	9770
Total magnetic moment	0.358 A m ⁻²	0.785	74,261
Refractive index	1.27	0.800	4764
Electronic conductivity	4.292×10 ⁵ S/m	0.882	23,218
Ionic conductivity	4.114×10 ⁵ S/m	0.893	23,218
p-Seebeck	45.295 μ V/K	0.758	23,218
n-Seebeck	40.042 μ V/K	0.609	23,218
Maximum piezoelectric e_{ij}	1.285 C/m ²	0.624	4799
Maximum piezoelectric d_{ij}	1.409 pC/N	0.660	3347
Maximum dielectric constant	1.297	0.683	4809
Maximum electronic dielectric constant	1.285	0.816	4809
Maximum ionic dielectric constant	1.581	0.797	4809

generalization ability was proved on an additional experimental dataset, with structures significantly exceeding the complexity of the training set. The Method LLM and Precursor LLM, fine-tuned on synthesis recipes from literature, achieve over 90% accuracy in classifying synthetic methods and identifying solid-state synthesis precursors for binary and ternary compounds, respectively. Furthermore, CSLLM identified 45,632 synthesizable theoretical structures from the MP database, with 23 key properties available for these structures.

However, accurately predicting synthesis conditions, such as temperature and pressure for any material, remains challenging due to insufficient data. This highlights the critical need for constructing high-quality datasets to fine-tune LLMs effectively. To address this need, a possible solution is leveraging the reinforcement learning with human feedback (RLHF) method⁶⁹, enabling iterative fine-tuning of the LLM through continuous feedback of experimental synthesis data. Despite these limitations, our CSLLM framework showcases the transformative potential of LLMs in bridging the gap between theoretical material design and practical synthesis. Its robust and powerful predictive capability, coupled with a user-friendly interface, renders it a valuable tool for researchers, enabling them to swiftly upload crystal structures and obtain reliable synthesizability and precursor prediction in a few seconds.

Methods

Fine-tuning and inference of LLMs⁷⁰

The Synthesizability LLM and Method, Precursors LLMs were fine-tuned using the pre-trained LLaMA3-8B⁷¹ models. For the Synthesizability LLM, we also tested LLaMA-7B³⁸, achieving a similar accuracy of 98.6%. All LLMs were fine-tuned utilizing the LoRA technique⁵⁹ and LMFlow⁷⁰ framework with $r=8$, $\alpha=32$, and a dropout rate of 0.1. LoRA was designed to efficiently adapt large pre-trained models to downstream tasks by introducing low-rank updates to the model's weights. This approach

significantly reduces the number of trainable parameters, making the fine-tuning process more computationally efficient while maintaining high performance. All the fine-tunings and inferences were performed on a server with one NVIDIA A800 GPU with 80 GB VRAM and 1 TB memory. Notably, the DeepSpeed library⁷² and FlashAttention⁷³ were used to reduce the GPU memory usage and improve training efficiency. The temperature parameter for all the inferences was set to 0, as higher values (e.g., 0.1 and 0.2) would reduce accuracy and increase the output of redundant information. The details about the loss function and parameters in fine-tuning and inference are shown in Supplementary Note 2-3, Table S2-4 and Fig S24-26.

Training of the CGTNet models

Among the 23 MPs, the HSE band gap data is sourced from the study by Yoon et al.⁷⁴, while the experimental band gap data is collected from the work of Li et al.⁷⁵. The PBE band gap and electrochemical stability (ΔG_{pbx}) are obtained from the MP database⁵¹, and the remaining properties are sourced from the JARVIS database⁵⁴. Simple properties include the mBJ band gap, PBE band gap, energy above hull, bulk modulus, shear modulus, Poisson's ratio, refractive index, electronic conductivity, ionic conductivity, maximum electronic dielectric constant, and maximum ionic dielectric constant. The remaining properties are classified as complex properties. For each of the 23 material property datasets, we trained our previously proposed transformer-based GNN model, CGTNet. This model has demonstrated better accuracy and small data friendliness compared to CGCNN⁷⁶, SchNet⁷⁷, PaiNN⁷⁸ DimeNet++⁷⁹, and GemNet⁸⁰. The main hyperparameters used for training CGTNets are shown in Supplementary Note 3.

GUI of CSLLM and material structure textualization

The GUI of CSLLM was built using the Gradio Python library. Uploaded CIF or POSCAR files are automatically converted to material strings. Specifically, the space group of the structure is first analyzed by the SpacegroupAnalyzer class in pymatgen⁸¹ with "symprec = 0.01" and "angle_tolerance = 5." Then, the structure is transformed into its primitive cell, and the Wyckoff positions are analyzed using the PyXtal library⁸². Finally, the material string undergoes inferences through the preloaded Synthesizability, Method, and Precursor LLM, followed by reaction energy calculations and other precursor recommendations. This entire process runs in the background, and the final output is presented as shown in Fig. 1a. For a video overview, see <https://github.com/szl666/CSLLM>.

Reporting summary

Further information on research design is available in the Nature Portfolio Reporting Summary linked to this article.

Data availability

The comprehensive list of all predicted properties of the synthesizable theoretical structures is provided in Supplementary Data 1 and Source Data files. Source data are provided with this paper. The initial non-synthesizable structures data used for fine-tuning the large language model has been deposited in Figshare under https://figshare.com/articles/dataset/Non-synthesizable_structures/29086931⁸³. The synthesizable structures data are obtained from the ICSD, which is subject to a commercial license and therefore cannot be publicly shared. Interested researchers may request access by contacting FIZ Karlsruhe directly through their official website (<https://www.fiz-karlsruhe.de/en/icsd>). Access is typically granted to individuals or institutions with a valid license for research purposes only.

Code availability

The codes to perform the CSLLM framework along with the result analysis and visualization are available at <https://github.com/szl666/CSLLM> under MIT license⁸⁴.

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Author contributions

Z.S., Q.Z. and J.W. conceived this work. Z.S. proposed the CSLLM framework and wrote the code with guidance from Q.Z., S.L. and J.W. Z.S.,

Q.Z., M.J. and J.W. analyzed the data and co-wrote the manuscript, with input from the other authors.

Competing interests

The authors declare no competing interests.

Additional information

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